Executive Analytics Report

SYNTHETIC DEMONSTRATION — NOT REAL COMPANY DATA

Scope and assumptions

This portfolio demonstrates an end-to-end e-commerce analytics workflow with a deterministic synthetic dataset (seed 27102026), 1,800 customers, 7,280 orders and 756 support tickets covering 2024–2025. It is a demonstration only, not actual company data. Recognized revenue includes only Completed orders with Successful payment. Recognized profit subtracts item cost and excludes overhead. Return rate is returned orders divided by all orders. On-time delivery rate is On time divided by On time + Late. Churn is a synthetic customer snapshot outcome.

Executive snapshot

The demo produces recognized revenue of $976,190 and recognized profit of $207,580; 6,098 completed/paid orders and AOV $160.08. Return rate is 7.1%; on-time rate is 83.9%; synthetic churn prevalence is 9.3%. These values should be used to review analytical workflow, not as a forecast or benchmark.

Three findings

1. Segment performance: Core is the highest-revenue segment ($436,351) and contributes $100,919 recognized profit. 2. Category economics: Electronics has the weakest observed contribution margin in this generated sample (7.6%; revenue $428,003, profit $32,396); margin excludes overhead and is sensitive to simulated cost assumptions. 3. Delivery/churn association: customers with any recorded late delivery show synthetic churn rate 11.7%, compared with 7.4% among customers without a late delivery. The relative rate is 1.57x; this is an association in generated data and is not causal evidence.

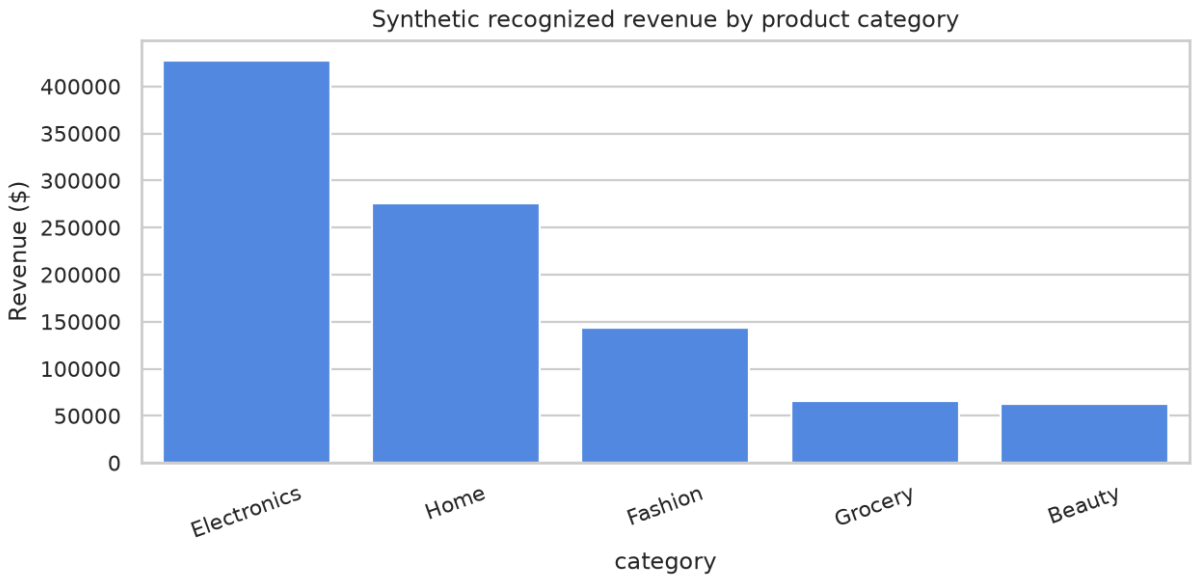


Figure: Synthetic recognized revenue by category. See notebook for full eight-chart EDA dashboard.

Operating signals

Payment method with the highest combined failed-or-refunded rate is Wallet (14.1%). City with the highest synthetic late-delivery rate is Kolkata (23.0%). Strongest/weakest monthly recognized revenue in the generated period is 2025-10 ($70,256) / 2024-02 ($20,713); seasonality in this small synthetic scenario is not a planning forecast. Refund/payment has the longest mean support resolution (4.6 days). The top-spending quartile contains 450 customers by rank-based quartile assignment.

Recommendations

1. Prioritize a measured delivery-reliability pilot in the city with the highest late rate. Randomize eligible shipments or use phased rollout, track on-time rate and customer churn/retention, and monitor cost and complaint guardrails; do not infer causality from the observed association. 2. Review unit economics for the weakest-margin category: validate landed cost, discounts, return handling and vendor terms; test price/assortment interventions before broad rollout. 3. Investigate the payment method with the highest failed/refunded rate; segment failures vs refunds, audit gateway/error codes, and run a controlled payment UX or routing test with conversion and fraud/refund guardrails.

Modeling, caveats, next steps

Useful candidate churn predictors include recency, order frequency, spend, change in order frequency, return/refund history, late-delivery exposure, support ticket volume and resolution time, payment failures, tenure, segment and city. Avoid leakage by fixing an observation window and predicting a later outcome window; evaluate time-based holdout, calibration, fairness, stability and business lift before operational use. The pipeline here is pedagogical and the synthetic label was generated using some correlated features, so its apparent predictive performance is not evidence of production model quality.

10-minute interview outline

0–1 min: state the business problem and the synthetic-data caveat. 1–2 min: show source tables, cleaning checks and metric definitions. 2–4 min: present category/segment and month charts. 4–5 min: explain the manual sample SD or Bayes calculation. 5–6 min: walk through the customer-revenue CTE/window SQL. 6–7 min: show RFM features and the leakage-aware pipeline. 7–8 min: discuss the late-delivery/churn association without causal language. 8–9 min: deliver three recommendations. 9–10 min: state limitations and next-week priorities (real data validation, true Power Query/PivotTable model, experiment design, time-based model validation).